

Power System Transient Stability Using Individual Machine Energy Functions

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Abstract—Recent efforts to apply direct methods of transient stability analysis to multimachine power systems have used the so-called “energy functions.” These functions describe the system transient energy causing the synchronous generators to depart from the initial equilibrium state, and the power network’s ability to absorb this energy so that the synchronous machines may reach a new post-disturbance equilibrium state. Recent results have shown that not all the transient energy contributes to system separation. Indeed it has become increasingly evident that system separation depends on the energy of certain individual machines or groups of machines, which comprise the *critical group*, and which tend to separate from the rest of the machines (which make up the *noncritical group*). Thus there is a need for generating energy functions for individual machines (or for groups of machines).

Using a center of inertia frame of reference, the energy function V_i for machine i is derived. A procedure for first swing transient stability assessment is developed using the energy function of individual machines and groups of machines. The method is tested extensively on two realistic power networks (the 20-generator IEEE System and the 17-generator reduced Iowa System).

An analytical justification for using the critical energy of individual machines in stability assessment is provided using an invariance principle for ordinary differential equations. Power system transient stability is analyzed using the energy function of the critical group of machines. This energy function is dependent on *all* state variables of the power system, and satisfies the hypotheses of the invariance theorem of La Salle, enabling us to deduce the asymptotic stability of the post-disturbance equilibrium of the entire power system. It also enables us to obtain an estimate of the domain of attraction of this equilibrium of the entire power system.

The methodology advanced herein, which combines computer-aided techniques with analytical tools, yields less conservative results than were obtained in previous works that used total system energy. It is to be noted that the present results are preliminary in the sense that the mechanism of separation of the critical group of machines from the rest of the system needs further investigation.

I. INTRODUCTION

IN THE last two decades numerous efforts have been made to apply direct methods of transient stability analysis to multimachine power systems. Of these, Lyapunov’s direct method which received considerable attention, has yielded conservative results (see review articles [1], [2]). More recently, success has been obtained with functions that describe the system transient energy and the power network’s ability to “absorb” this energy so that the synchronous machines may reach a new post-disturbance

equilibrium state. Some energy functions describe the system transient energy using a synchronous frame of reference [3]–[5]. Others, as in the approach proposed here, have used a center of inertia (COI) formulation [6]–[9]. Success of the techniques recently used in first swing direct stability analysis can be attributed to progress in two main areas:

- 1) More accurate determination of the region of stability, especially identifying the unstable equilibrium point (u.e.p.) “relevant” to the post-disturbance trajectory.
- 2) Better accounting of the system transient energy, especially the energy causing separation of one or more machines from the system, and where it resides.

In spite of the recent successes achieved with the use of energy functions, difficulties in each of the above two areas have remained to be resolved. In the first area, situations are often encountered in power systems where complex “modes” of instability arise, making it difficult to determine (with confidence) the relevant u.e.p. In the second area it was recently discovered [8], [9] that significant portions of the system transient energy do not contribute to system separation; hence it should be corrected for in transient stability assessment.

In the recent work with which the authors have been associated, the above difficulties were resolved by “pragmatic” means, i.e., using engineering judgment. However, it was suspected that the problem was more fundamental than should be resolved by making “corrections” in the expression for the system energy function. From the numerous simulations conducted by the authors it was becoming increasingly evident that system separation does *not* depend on the total system energy, but rather on the transient energy of individual machines or groups of machines tending to separate from the rest. It was reasoned, therefore, that a new approach was needed. Such an approach should be cognizant of the physical behavior of the machines in the post-disturbance period, as revealed by simulation studies, supported by the proper theory. This is the central issue dealt with in this paper. The approach, which is developed and tested here, uses the energy function for individual machines. The theoretical justification is based on an invariance principle for ordinary differential equations.

To help the reader who is not familiar with power system dynamics, a few remarks concerning power system transient stability are in order.

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Power System Transient Stability

Transient instability in a power system is caused by a severe disturbance which creates substantial imbalance between the input power supplied to the synchronous generators (by the prime-movers) and their electrical power outputs. Some of the severely disturbed generators will "swing" far enough from their equilibrium positions losing synchronism in the process. Such a severe disturbance may be a sudden and large change in load, generation, or network configuration. Usually the severe disturbance under which transient stability is tested, is a short circuit in the high-voltage transmission network. In power system terminology this is referred to as a *fault*.

When a fault occurs, certain generators which are electrically close to the fault location are disturbed to a greater extent than the other generators which are remote from the fault. These generators tend to accelerate or decelerate, depending on the nature of the fault, from the rest of the generators in the system. If the fault lasts long enough, eventually one machine or a group of machines separate from the rest of the system, causing instability (loss of synchronism). However, the power network is equipped with automatic devices that sense the existence of the faults in the network and initiate action to "clear" the fault, i.e., isolate the faulted section of the network. A matter of great practical importance, therefore, is that the time required (by the protective equipment) to clear the fault should be less than the duration of the fault that would create a disturbance large enough to cause one or more machines to lose synchronism. This is the so-called *critical clearing time*, commonly used in the literature on transient stability of power systems. This term is often quoted by researchers on power system transient stability by direct methods as a figure of merit to be used: a) to compare results obtained by time simulation with those obtained by direct methods, and b) to determine how "robust" a power network may be as it is subjected to disturbance. (The state vector, evaluated at the critical clearing time, provides a means of estimating the region of attraction of the post-fault system. This will be demonstrated in Section VII.)

Focusing our attention on the machine or group of machines tending to separate from the rest, we will call this group "the critical group." The rest of the system, which includes the majority of the synchronous generators, forms the "non-critical group." It is important to note that the machines in the noncritical group remain in synchronism among themselves in the initial phase of the transient, which is the phase covered by the transient stability investigation under consideration here. Hence, in reality, power system transient stability deals with the transient behavior of a small group of generators with respect to the remainder of the generators. Thus the stability (or instability) of the critical group is what determines the transient stability of the overall system.

Scope of the Paper

In this paper, the authors aim to develop energy functions for individual machines (or for groups of machines);

investigate the region of attraction of the equilibrium of the entire post-fault system, using this energy function; compare this assessment with results obtained by digital simulation; and provide a theoretical justification for the stability assessment based on individual machine energy. This work provides a departure from the state of the art of the last two decades [1], [2] in analyzing power system stability. Instead of assessing stability from a system-wide viewpoint, we concentrate on the stability of the critical group of machines, knowing that the noncritical group maintains synchronism.

The approach used in this paper combines computer-aided simulation, physical intuition, and heuristic arguments with sound engineering judgment. These features are then supplemented by a qualitative analysis to establish sharper stability results than were obtained in previous works that used total system energy [6]–[9]. It should be noted that the results presented in this paper are preliminary in nature, since the mechanism of separation of the critical group from the rest of the system needs further investigation.

II. POWER SYSTEM MODEL

Consider a multi-machine power system. In this case the power network is represented by the so-called classical model [10, ch. 2], i.e., by generators which are represented by constant voltage behind transient reactance, constant impedance loads, etc. For this model the motion of the generators is governed by the set of differential equations

$$\left. \begin{aligned} M_i \dot{\omega}_i + D_i \omega_i &= P_i - P_{ei} \\ \dot{\delta}_i &= \omega_i, \quad i = 1, 2, \dots, n \end{aligned} \right\} \quad (1)$$

where for machine i ,

- δ_i angle of voltage behind transient reactance, indicative of generator rotor position,
- ω_i rotor speed,
- M_i generator inertia constant,
- D_i damping coefficient.

The expressions for P_i and P_{ei} are given by

$$\left. \begin{aligned} P_i &= P_{mi} - E_i^2 G_{ii} \\ P_{ei} &= \sum_{j=1}^n [C_{ij} \sin(\delta_i - \delta_j) + D_{ij} \cos(\delta_i - \delta_j)] \end{aligned} \right\} \quad (2)$$

where,

$$C_{ij} = E_i E_j B_{ij}$$

$$D_{ij} = E_i E_j G_{ij}$$

- P_{mi} mechanical power input,
- E_i magnitude of voltage behind transient reactance,
- G_{ii} real part of the i th diagonal element of the network's Y -matrix,
- G_{ij}, B_{ij} real and imaginary components of the ij th element of the network's Y -matrix.

Good physical insight into the behavior of the synchronous generators is obtained when their motion is expressed with respect to the machines' COI [6]–[9]. The position of the COI is defined by

$$\delta_0 = \frac{1}{M_T} \sum_{i=1}^n M_i \delta_i \quad (3)$$

where

$$M_T = \sum_{i=1}^n M_i.$$

The motion of the COI is determined by

$$\begin{aligned} M_T \dot{\omega}_0 &= - \sum_{i=1}^n D_i \omega_i + \sum_{i=1}^n (P_i - P_{ei}) \\ &= - \sum_{i=1}^n D_i \omega_i + P_{\text{COI}} \\ \dot{\delta}_0 &= \omega_0 \end{aligned} \quad (4)$$

where

$$P_{\text{COI}} = \sum_{i=1}^n (P_i - P_{ei}).$$

The generator's motion with respect to the COI can now be obtained. First we define θ_i and $\tilde{\omega}_i$ by

$$\theta_i = \delta_i - \delta_0, \quad \tilde{\omega}_i = \dot{\theta}_i = \omega_i - \omega_0. \quad (5)$$

Since the motion of the individual generators is damped, the motion of their inertial center will also be damped. We can assume that the COI has an effective damping coefficient equal to D_0 by the following reasoning. For the case $\omega_1 = \omega_2 = \dots = \omega_n \neq 0$,

$$D_0 = \sum_{i=1}^n D_i > 0.$$

In general, D_0 is a function of ω_i , however, we will assume that the variation of D_0 with respect to ω_i is small and is negligible. Then (4) becomes

$$M_T \dot{\omega}_0 = -D_0 \omega_0 + P_{\text{COI}} \quad (6)$$

where D_0 is defined by

$$D_0 \omega_0 = \sum_{i=1}^n D_i \omega_i \quad (7a)$$

or

$$D_0 \sum_{i=1}^n \frac{M_i}{M_T} \omega_i = \sum_{i=1}^n D_i \omega_i. \quad (7b)$$

From (1) and (6) the equations of motion of the generators in the COI reference frame are now given by

$$\left. \begin{aligned} M_i \ddot{\theta}_i &= -D_i \tilde{\omega}_i - \left(D_i - \frac{M_i}{M_T} D_0 \right) \omega_0 \\ &\quad + P_i - P_{ei} - \frac{M_i}{M_T} P_{\text{COI}} \\ \dot{\theta}_i &= \tilde{\omega}_i, \quad i = 1, \dots, n. \end{aligned} \right\} \quad (8)$$

We note that the contributions of machine i to the damping and to the acceleration of the COI are given by the terms $(D_i - (M_i/M_T)D_0)$ and $((M_i/M_T)P_{\text{COI}})$, respectively.

It should be pointed out that when power system stability analysis is conducted using either the model described by (1) or (8), it is customary to neglect damping. The reasoning is that this model is adequate only for the so-called first swing transient. The peak of this first swing is reached typically in less than 0.5 s. In this period damping (usually, the damping coefficient D_i is in the order of 1.0 per unit) has negligible effect on the magnitude of the swing. When synchronism is maintained in the first swing, it is assumed that damping will reduce the subsequent swings. However, damping is included in the present treatment for completeness.

We conclude by noting that for the special case of *uniform damping* (i.e., when $D_i/M_i = \text{constant} = D$, $i = 1, 2, \dots, n$), we can show that the equations of motion of the generators assume the form

$$\left. \begin{aligned} M_i \ddot{\theta}_i &= -D_i \tilde{\omega}_i + P_i - P_{ei} - \frac{M_i}{M_T} P_{\text{COI}} \\ \dot{\theta}_i &= \tilde{\omega}_i, \quad i = 1, 2, \dots, n. \end{aligned} \right\} \quad (9)$$

Furthermore, when *damping is neglected*, the system motion is described by the equations

$$\left. \begin{aligned} M_i \ddot{\theta}_i &= P_i - P_{ei} - \frac{M_i}{M_T} P_{\text{COI}} \\ \dot{\theta}_i &= \tilde{\omega}_i, \quad i = 1, 2, \dots, n. \end{aligned} \right\} \quad (10)$$

III. TRANSIENT ENERGY OF INDIVIDUAL MACHINES

In this section, we develop an expression for the individual machine transient energy. In this derivation, the effect of damping is neglected in the system description because the energy function is used to examine first swing stability. In that period, damping has negligible effect on the magnitude of the swing, and acts in the direction to enhance stability. This is illustrated in the data shown in Section VI, Tables II and III. With this objective in mind, we proceed as follows. Multiplying the i th post-fault swing equation by $\dot{\theta}_i$ and rearranging, we obtain the expression (with $D_i = 0$)

$$\left[M_i \dot{\theta}_i - P_i + P_{ei} + \frac{M_i}{M_T} P_{\text{COI}} \right] \dot{\theta}_i, \quad i = 1, 2, \dots, n. \quad (11)$$

Integrating (11) with respect to time, using as a lower limit $t = t_s$, where $\tilde{\omega}(t_s) = 0$ and $\theta(t_s) = \theta^s$ is the stable equilibrium point (s.e.p.), yields

$$\begin{aligned} V_i &= \frac{1}{2} M_i \tilde{\omega}_i^2 - P_i (\theta_i - \theta_i^s) + \sum_{j=1, j \neq i}^n C_{ij} \int_{\theta_i^s}^{\theta_i} \sin \theta_{ij} d\theta_{ij} \\ &\quad + \sum_{j=1, j \neq i}^n D_{ij} \int_{\theta_i^s}^{\theta_i} \cos \theta_{ij} d\theta_{ij} + \frac{M_i}{M_T} \int_{\theta_i^s}^{\theta_i} P_{\text{COI}} d\theta_i, \\ &\quad i = 1, 2, \dots, n. \end{aligned} \quad (12)$$

This integral is evaluated using the values of θ_i , θ_j , $\tilde{\omega}_i$ as obtained from the solution of (8) and using the values C_{ij} , D_{ij} , and θ_i^s for the post-disturbance network. The first term in the right-hand side of (12) is the kinetic energy of machine i with respect to the system inertial center. It is customary to consider the remaining terms as a potential energy [6]–[9]. Thus (12) can be expressed as

$$V_i = V_{KEi} + V_{PEi}. \quad (13)$$

Expression for $\dot{V}_i$

In the above approach, we arrived at the expression (12) for V_i by *neglecting damping*. In the following, we evaluate the derivative of V_i with respect to t along the trajectories of system (8), $\dot{V}_{i(8)}$ with *damping included* in the system equations. This is completely consistent with the Lyapunov approach: given V_i (regardless of how it was obtained), we need to determine the properties of $\dot{V}_i$ with respect to the system in question.

From (11) we obtain

$$V_i = \frac{1}{2} M_i \tilde{\omega}_i^2 + \int_{\theta_i^s}^{\theta_i} \left(-P_i + P_{ei} + \frac{M_i}{M_T} P_{COI} \right) d\theta_i, \quad i = 1, 2, \dots, n. \quad (14)$$

Differentiating with respect to time, we obtain

$$\begin{aligned} \dot{V}_i &= M_i \tilde{\omega}_i \dot{\tilde{\omega}}_i - \left(P_i - P_{ei} - \frac{M_i}{M_T} P_{COI} \right) \dot{\theta}_i \\ &= \tilde{\omega}_i \left[M_i \dot{\tilde{\omega}}_i - \left(P_i - P_{ei} - \frac{M_i}{M_T} P_{COI} \right) \right], \quad i = 1, 2, \dots, n. \end{aligned} \quad (15)$$

From (8) and (15) we have

$$\dot{V}_i = \tilde{\omega}_i \left[-D_i \tilde{\omega}_i - \left(D_i - \frac{M_i}{M_T} D_0 \right) \omega_0 \right], \quad i = 1, 2, \dots, n. \quad (16)$$

Similarly, we can show that in the *special case of uniform damping* we have

$$\dot{V}_i = -D_i \tilde{\omega}_i^2, \quad i = 1, 2, \dots, n \quad (17)$$

and when damping is neglected, we have

$$\dot{V}_i = 0, \quad i = 1, 2, \dots, n. \quad (18)$$

Sign Definiteness of V_i and $\dot{V}_i$

Since (12) contains path-dependent integrals, the sign definiteness of V_i cannot readily be determined analytically in closed form. It has always been conjectured that V_i is positive definite (see [6]–[9] and [11, pp. 124–125]). Extensive simulations in [6]–[9], as well as by the present authors, support the argument that V_i is positive definite.

We can easily show that (16) can be expressed as

$$\dot{V}_i = -D_i (\omega_i - \omega_0)^2 - \left(D_i - \frac{M_i}{M_T} D_0 \right) \omega_0 (\omega_i - \omega_0). \quad (19)$$

The first term on the right-hand side of (19) is obviously negative, except at the stable equilibrium point where it is zero. For the second term on the right-hand side, again

extensive simulations indicate that $\dot{V}_i$ is negative except at the origin. Intuitively, it appears reasonable to expect $\dot{V}_i < 0$ except at the origin. Consider the case where the disturbance injects transient energy into the system making $\omega_0 > 0$ (except when there is an infinite inertia in the system). The severely disturbed (i.e., critical) machines diverge from the inertial center (i.e., they accelerate away from the inertial center), and these machines in general have $\omega_i - \omega_0 > 0$. Thus if i denotes a critical machine, $(\omega_i - \omega_0) \geq 0$. Since for this machine $(D_i - (M_i/M_T) D_0) > 0$, and $\omega_0 > 0$, the second term on the right-hand side of (19) is negative, except at the stable equilibrium point, where it is equal to zero. Thus for the *critical* machines, $\dot{V}_i$ is negative semidefinite.

IV. SOME CONJECTURES SUPPORTED BY SIMULATIONS

A. Critical Energy of Individual Machines

The transient energy, injected into the system during the disturbance, increases V_i and causes machine i to diverge from the rest of the system. When the disturbance is removed, and as a machine i continues to diverge from the rest of the system, its kinetic energy is being converted into potential energy. This motion will continue until the initial kinetic energy is totally converted to potential energy. When this takes place, the machine will converge toward the rest of the system. The presence of damping will ensure that stable equilibrium conditions will eventually be reached. On the other hand, if the kinetic energy is not totally converted to potential energy, machine i will lose synchronism with the system. Since the potential energy varies along the post-disturbance trajectory, a number of questions immediately present themselves: Does it have a maximum value along the trajectory? Does this maximum value (if it exists) vary with different disturbances resulting in different modes of instability? How can this value be determined? We answer these questions by arguments derived from the determination of the region of stability for the *total* system energy functions. An excellent treatment of this subject is given in reference [4] for energy functions using the synchronous frame of reference. The work in [4] was extended to total system energy using COI reference frame in [7], [9].

By making use of *total* system energy functions, part of the boundary of the region of stability is determined by hypersurfaces which are orthogonal to the equipotential surfaces, and which pass through the saddle points. These hypersurfaces form the principal energy boundary surface (PEBS). At the PEBS, the potential energy for the system is maximum, and on the boundary of the region of stability, the total energy of the system is such that it is equal in magnitude to the potential energy at the PEBS. Near a particular unstable equilibrium point, the potential energy is fairly “flat”. Since at the PEBS the potential energy is maximum, stability is maintained if the *system* kinetic energy is totally converted to potential energy before reaching the PEBS, and the converse is also true [4], [7], [9].

We propose to extend the above concepts to the individual machine energy function V_i . Thus we suggest that for machine i to remain stable, the system has to have a region of attraction containing the hypersurfaces where V_{PEi} is maximum, and this maximum value of V_{PEi} is fairly "flat." Also, we suggest that $V_{i/critical}$, the critical value of the total energy of machine i at the instant of clearing (which provides an estimate of the region of attraction) is equal in magnitude to $V_{PEi/max}$. We have come to this conclusion after originally observing that the behavior of the total system energy function is dominated by the behavior of the energy of the critical machines [8], [9], [12]. It is reasonable, therefore, to expect similar characteristics of the region of stability for the energy function describing them. This assumption has been fully supported by the results obtained.

In the present work, numerous simulations have been conducted on three systems. It was found that if the fault is kept long enough for one machine (or more) to become critically unstable, the potential energy of the critical machine goes through a maximum before instability occurs. Furthermore, this maximum value (of the potential energy along the post-disturbance trajectory) of a given machine is essentially independent of the duration of the disturbance. This value of potential energy represents the energy absorbing capacity of the network for machine i and hence is the critical value of V_i , i.e.,

$$V_{i/critical} = V_{PEi/max} \text{ along trajectory.}$$

Finally, an analytical justification of this assumption is given in Section VII.

B. Determination of the Mode of Instability

It has been established in the previous section that, for a given disturbance location and post-disturbance network, the value of $V_{critical}$ for a given machine is essentially independent of the duration of the disturbance. Therefore, a convenient method to determine the mode of instability is provided by examining the sustained fault trajectory.

By simulating a sustained fault (or a fault of long duration), the potential energy terms of (12) are computed (V_{PEi} , $i = 1, 2, \dots, n$) for various instants. The values of $V_{PEi/max}$ are noted for the different machines (or groups of machines). These are the same as $V_{critical}$, i.e.,

$$V_{i/critical} = V_{PEi/max}. \quad (20)$$

Instability occurs if, at the instant of fault clearing, the total transient energy of the critical machines exceeds their critical energy.

C. Calculation of $V_{critical}$ for Individual Machines

Examining (12) we note that the integrands are not independent of the trajectory. Therefore, the individual machine energies, along the faulted trajectory, are evaluated numerically. A special computer program, using the trapezoidal rule for integration, has been written for this purpose. This program has been used in the simulations presented later in this paper.

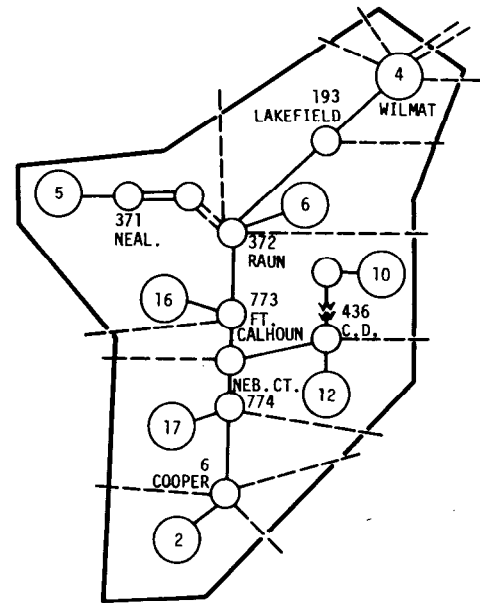


Fig. 1. Area of interest 17-generator system.

V. TRANSIENT STABILITY ASSESSMENT USING THE INDIVIDUAL MACHINE ENERGY

The procedure for transient stability assessment using the transient energy of individual machines (or groups of machines) is outlined below.

- Step 1: For the post-disturbance network, the stable equilibrium point θ^s and the short-circuit admittance matrix Y_{BUS} are determined.
- Step 2: For each candidate fault location, a sustained fault case is run. The values V_i , $i = 1, 2, \dots, n$, are computed along the faulted trajectory using the special computer program.
- Step 3: By examining V_i and its components, the values of $V_{i/critical} = V_{PEi/max}$ are determined and stored for each fault location.
- Step 4: For a given disturbance, the values of θ_i and $\bar{\omega}_i$ are obtained at the end of the disturbance, i.e., at fault clearing. From this information $V_{i/t=t_c}$ is computed.
- Step 5: Transient stability is checked. Machine i will be stable if $V_{i/t=t_c} < V_{i/critical}$, and unstable if $V_{i/t=t_c} > V_{i/critical}$.

VI. SIMULATION STUDIES

A. Test Networks

The procedure for assessing transient stability of a multi-machine power system using the transient energy of individual machines was tested by extensive simulation on two moderately large power systems described below.

1) A 17-generator 163-bus system which is a reduced version of the network of the State of Iowa. This network is described in [8], [9]. The disturbances investigated are three-phase faults at four generating stations in the western part of the state. Fig. 1 shows only the main high-voltage lines interconnecting the generating stations.

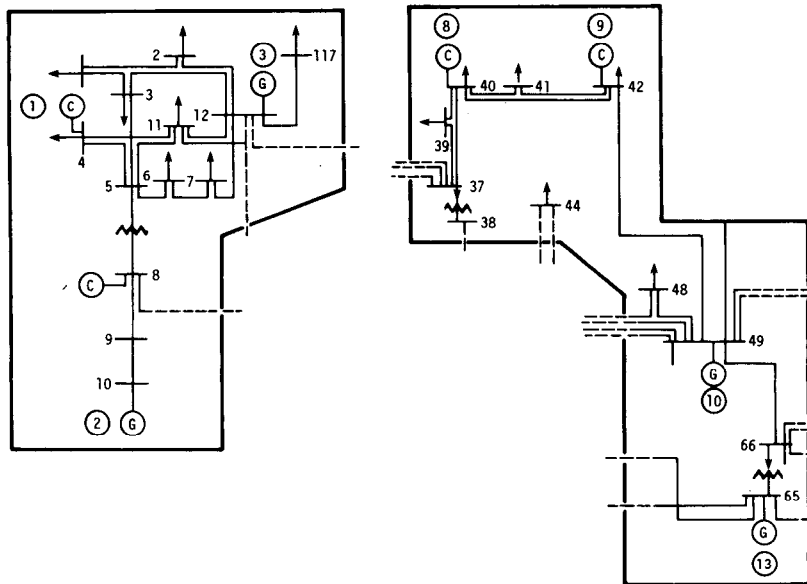


Fig. 2. Areas of special interest IEEE 118-bus system.

2) The IEEE 20-generator 118-bus system which was investigated in [7]. This system is described in [13]. Three-phase faults are applied at seven locations, near the terminals of generators or synchronous condensers. In all cases, the fault is cleared without line switching (to compare results with those published in [7]). Two areas of special interest in that network are shown in Fig. 2.

B. Critical Transient Energy

For the faults investigated, the maximum potential energy of the critical machines is computed for the critically unstable condition and for the sustained fault. We note that in some cases additional machines lose synchronism with the sustained fault. In each case, V_{critical} for the machine or machines initially losing synchronism is given in Table I.

The data in Table I show that for all the cases investigated in both systems, the maximum potential energy for the critical machines is obtained for two trajectories: one at which the critical machines barely become unstable, and one for the sustained fault. In all cases, the value of $V_{\text{PE}/\text{max}}$ is fairly constant, even when additional machines become unstable with the sustained fault.

Thus, the conjecture that $V_{i|\text{critical}}$ for a given disturbance is fairly constant has been verified by simulation.

C. Stability Assessment by Individual Machine Energy

Simulations of several disturbances on each system were conducted. For each disturbance, two cases were run: a critically stable case and a critically unstable case. (In the latter case, the critical machines barely maintain or lose synchronism with the system.) At the instant the disturbance is removed (i.e., $t = t_c$), the value of V_i is computed. Stability assessment based on $V_{i|\text{critical}}$ is made and compared to the results obtained by simulation. The results are displayed in Tables II–V.

Table II shows the results for four disturbances in the Iowa System, first with damping neglected and with damping included ($D_i = 1$ pu). In every case, the critical clearing time based on $V_{i|t=t_c} = V_{i|\text{critical}}$ checks well with the data obtained by simulation, i.e., the critical clearing time falls between the critically stable and the critically unstable clearing times. Furthermore, the transient stability assessment yields the correct prediction of stability (or instability). The data in Table II also support the point made earlier in Section II concerning the effect of damping on first swing stability. In the four cases studied, essentially the same critical clearing times are obtained with and without damping; also, damping tends to slightly enhance stability.

Table III shows the results for five disturbances on the IEEE system that were investigated in [7]. Again, there is good agreement between stability assessment using $V_{i|\text{critical}}$ and with that obtained by time solution. Furthermore, the values of critical clearing times are generally less conservative than those obtained in [7], as can be seen from the comparison displayed in Table IV.

Three disturbances are of particular interest since the machines initially losing synchronism are *different* from the machines at which the fault is applied, these cases, presented separately in Table V are:

- The Reduced Iowa System, where the initial conditions are altered so that 200 MW of generation are shifted from Gen. #4 (Wilmarth) to Gen. #6 (Raun). With the fault applied at the Ft. Calhoun terminal (Gen. #16), the Raun generators (#5,6) first lose synchronism, while Gen. #16 does not.
- The IEEE System, with the same initial conditions. A fault at the terminal of Gen. #3 causes Gen. #2 to lose synchronism first; and with a fault at the terminal of Gen. #13, the first generators that become unstable are Gen. #8,9.

TABLE I
CRITICAL TRANSIENT ENERGY FOR CRITICAL MACHINES

Fault Location	Critically Unstable Case			Sustained Fault	
	t_c s	Critical Machines #	$V_{critical}$ (pu)	Unstable Machines #	$V_{critical}^*$ (pu)
<u>Reduced Iowa System</u>					
Raun	0.1924	5,6	$V(5,6) = 18.999$	5,6	$V(5,6) = 18.4312$
C.B. #3	0.204	12	$V(12) = 11.808$	2,5,6,10, 12,16,17	$V(12) = 11.5305$
Ft. Calhoun	0.354	16	$V(16) = 12.788$	2,5,6,10, 12,16,17	$V(16) = 12.2501$
Cooper	0.211	2	$V(2) = 11.103$	2,5,6,10, 12,16,17	$V(2) = 11.0437$
<u>IEEE System</u>					
Gen. #2	0.190	2	$V(2) = 5.026$	2	$V(2) = 4.9565$
Gen. #3	0.480	2	$V(2) = 5.305$	2,3	$V(2) = 5.2510$
Gen. #4	0.340	4,5	$V(4,5) = 7.686$	4,5	$V(4,5) = 7.5932$
Gen. #5	0.400	4,5	$V(4,5) = 11.886$	4,5	$V(4,5) = 11.720$
Gen. #9	0.480	9	$V(9) = 4.603$	9	$V(9) = 4.5988$
Gen. #13	0.340	8,9	$V(8,9) = 1.559$	1,2,3,4,5, 6,7,8,9, 10,11,12, 13,14,20	$V(8,9) = 1.5532$
Gen. #18	0.360	18	$V(18) = 4.474$	17,18	$V(18) = 4.4710$

*.Maximum potential energy (along the sustained fault trajectory) for the group of machines that first becomes unstable.

TABLE II
STABILITY ASSESSMENT USING INDIVIDUAL MACHINE ENERGY

Fault Location	Critical Machines #	Critically Stable Case		Critically Unstable Case		Assessment Based on V_{cri}		Critical Clearing time-s
		t_c s	V_{total} pu	t_c s	V_{total} pu	$V_{critical}$ pu		
Reduced Iowa System (Damping Neglected)								
Raun	5,6	0.192	$V(5) = 1.6227$	0.1924	$V(5) = 1.8248$	$V(5) = 1.6233$	0.1920	
			$V(6) = 18.0393$		$V(6) = 18.3828$	$V(6) = 18.3309$	0.1923	
			$V(5,6) = 17.2026$		$V(5,6) = 18.9997$	$V(5,6) = 18.4312$	0.1923	
C.B. #3	12	0.200	$V(12) = 11.0809$	0.204	$V(12) = 11.8321$	$V(12) = 11.5305$	0.202	
Ft. Calhoun	16	0.345	$V(16) = 11.8768$	0.356	$V(16) = 12.7942$	$V(16) = 12.2501$	0.350	
Cooper	2	0.204	$V(2) = 9.9535$	0.212	$V(2) = 11.2988$	$V(2) = 11.0437$	0.210	
Reduced Iowa System (Damping Included)								
Raun	5,6	0.192	$V(5) = 1.5526$	0.1924	$V(5) = 1.7993$	$V(5) = 1.5949$	0.1921	
			$V(6) = 17.5967$		$V(6) = 18.3995$	$V(6) = 18.3249$	0.1923	
			$V(5,6) = 16.6721$		$V(5,6) = 18.8476$	$V(5,6) = 18.3994$	0.1923	
C.B. #3	12	0.200	$V(12) = 11.0345$	0.204	$V(12) = 11.7534$	$V(12) = 11.4725$	0.202	
Ft. Calhoun	16	0.345	$V(16) = 11.7325$	0.356	$V(16) = 12.7670$	$V(16) = 12.1309$	0.350	
Cooper	2	0.204	$V(2) = 9.8258$	0.212	$V(2) = 11.2714$	$V(2) = 11.0213$	0.209	

Again, the data in Table V show that with the use of $V_{critical}$ for individual machines, the correct mode of instability is determined, and the critical clearing time for the critical machines can be predicted with the same accuracy obtained by time simulation. For the special cases of the IEEE system, the results in Table V match the time simulation more accurately than those in [7].

VII. ANALYTICAL JUSTIFICATION OF STABILITY ASSESSMENT USING INDIVIDUAL MACHINE ENERGY

In this section, we give a mathematical analysis for the stability assessment procedure using individual machine energy. We also provide a means of estimating the domain

TABLE III
STABILITY ASSESSMENT USING INDIVIDUAL MACHINE ENERGY

		Critically Stable Case		Critically Unstable Case		Assessment Based on V_{cri}	
Fault Location	Critical Machines #	t_c^s	V_{total} pu	t_c^s	V_{total} pu	$V_{critical}$ pu	Critical Clearing time-s
IEEE System							
Gen. #2	2	0.180	$V(2) = 4.4653$	0.190	$V(2) = 5.2415$	$V(2) = 4.9565$	0.186
Gen. #4	4,5	0.320	$V(4) = 8.2416$	0.340	$V(4) = 9.8113$	$V(4) = 9.7834$	0.340
			$V(5) = 1.9624$		$V(5) = 2.1975$	$V(5) = 2.0284$	0.3261
			$V(4,5) = 6.4312$		$V(4,5) = 7.6803$	$V(4,5) = 7.5932$	
Gen. #5	4,5	0.380	$V(4) = 1.3777$	0.400	$V(4) = 1.6322$	$V(4) = 1.4292$	0.384
			$V(5) = 12.1356$		$V(5) = 13.7196$	$V(5) = 13.3751$	0.396
			$V(4,5) = 10.3421$		$V(4,5) = 11.8870$	$V(4,5) = 11.720$	0.398
Gen. #9	9	0.460	$V(9) = 3.4310$	0.480	$V(9) = 4.6393$	$V(9) = 4.5988$	0.479
Gen. #18	18	0.340	$V(18) = 4.0335$	0.360	$V(18) = 4.5320$	$V(18) = 4.4710$	0.358

TABLE IV
COMPARISON OF CRITICAL CLEARING TIMES IEEE SYSTEM

Fault Location	Critical Clearing times-s		
	Using Individual Machine Energy	Obtained by Time Solution	Obtained in Ref. [7]
IEEE SYSTEM			
Gen. #2	0.186	$0.180 < t_{cr} < 0.190$	0.180
Gen. #4	0.339	$0.320 < t_{cr} < 0.340$	0.310
Gen. #5	0.398	$0.380 < t_{cr} < 0.400$	0.360
Gen. #9	0.479	$0.460 < t_{cr} < 0.480$	0.460
Gen. #18	0.358	$0.340 < t_{cr} < 0.360$	0.330

TABLE V
STABILITY ASSESSMENT USING INDIVIDUAL MACHINE ENERGY:
SPECIAL CASES

		<u>Critically Stable Case</u>		<u>Critically Unstable Case</u>		<u>Assessment Based on V_{cri}</u>	
<u>Fault Location</u>	<u>Critical Machines #</u>	t_c s	V_{total} pu	t_c s	V_{total} pu	$V_{critical}$ pu	<u>Critical Clearing time-s</u>
<u>Reduced Iowa System</u>							
Ft. Calhoun (Gen. #16)	5,6	0.310	V(5) = 0.8884	0.3314	V(5) = 1.1105	V(5) = 0.9972	0.321
			V(6) = 6.6099		V(6) = 7.1924	V(6) = 7.0145	0.325
			V(5,6) = 6.5213		V(5,6) = 7.0131	V(5,6) = 6.9214	0.327
			V(16) = 6.5695		V(16) = 9.4860	V(16) = 11.7178	
<u>IEEE System</u>							
Gen. #3	2	0.460	V(2) = 3.7016 V(3) = 7.2902	0.480	V(2) = 5.3691 V(3) = 7.6281	V(2) = 5.2510 V(3) = 9.2627	0.479
Gen. #13	8,9	0.320	V(8) = 0.7313 V(9) = 0.6746 V(8,9) = 1.3621 V(13) = 12.9372	0.340	V(8) = 0.8135 V(9) = 0.7544 V(8,9) = 0.5599 V(13) = 15.5678	V(8) = 0.8021 V(9) = 0.7470 V(8,9) = 1.5532 V(13) = 17.0606	0.336 0.338 0.339

of attraction of the stable post-fault equilibrium, by making use of the system state variables evaluated at the critical clearing time (obtained in the previous section), by making use of the individual machine energy functions, and by invoking an invariance principle. For the sake of completeness, we first provide the required background material for the invariance principle.

A. An Invariance Theorem

Let V and W be arbitrary sets. Then $V \times W$ denotes the cartesian product of V and W . If V is a subset of W , we write $V \subset W$ and if x is an element of V , we write $x \in V$. We let x^T denote the transpose of x . If d is a function or mapping of V into W , we write $d: V \rightarrow W$. Let R^n be n -dimensional real space and let $|\cdot|$ denote any one of the equivalent norms on R^n . Let R denote the real numbers and let $R^+ = [0, \infty)$. We say that a continuous function $\psi: [0, r_1] \rightarrow R^+$ for some $r_1 > 0$, resp., $\psi: R^+ \rightarrow R^+$, belongs to class K (i.e., $\psi \in K$), if

- (i) $\psi(0) = 0$, and
- (ii) $\psi(r_1) > \psi(r_2)$ whenever $r_1 > r_2$.

We consider systems described by ordinary differential equations of the form

$$\begin{cases} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{cases} \quad (21a)$$

where $\dot{x} = dx/dt$, $t \in R^+$, $x \in R^n$, $y \in R^m$, $f: R^n \times R^m \rightarrow R^n$, and $g: R^n \times R^m \rightarrow R^m$. If we let $(x^T, y^T) = z^T \in R^{n+m}$ and $h^T(z) = (f^T(x, y), g^T(x, y))$, then (21a) can equivalently be written as

$$\dot{z} = h(z). \quad (21b)$$

We assume that f and g are sufficiently smooth to ensure for every $t_0 \geq 0$ and for every $z_0 \in R^{n+m}$ the existence of unique solutions $z(t; z_0, t_0)^T = (x(t; z_0, t_0)^T, y(t; z_0, t_0)^T)$ for (21), which exist for all $t \geq t_0$, where $z(t_0; z_0, t_0) = z_0$. Furthermore, we assume that $z = 0$ is an isolated equilibrium of (21b). Thus $f(0, 0) = 0$ and $g(0, 0) = 0$.

In the following result, which is a version of the La Salle invariance theorem, we require the existence of a continuously differentiable function $V: D \rightarrow R$, where D is a domain in R^{n+m} containing the origin $z = 0$. Recall that the derivative of V with respect to t along the solutions of (21b) is given by

$$\dot{V}_{(21b)}(x, y) = \dot{V}_{(21b)}(z) = \nabla V^T(z) h(z)$$

where ∇V denotes the gradient vector of V .

Theorem 1

Suppose there exists a continuously differentiable function $V: D \rightarrow R$ and a function $\psi \in K$ such that

- (i) $V(x, y) = V(z) \geq \psi(|z|)$, $V(0, 0) = 0$, and
- (ii) $\dot{V}_{(21b)}(x, y) = \dot{V}_{(21b)}(z) \leq 0$

for all $(x, y) \in D$. For some real constant $k > 0$, let H_k be that component of the set $S_k = \{z \in R^{n+m}: V(z) \leq k\}$ which contains the origin $z = 0$. Suppose that $k > 0$ is sufficiently small so that H_k is a closed and bounded subset of D . (Such a k will always exist since in view of (i), V is positive

definite (see [14])). Let

$$E = \{z \in H_k: \dot{V}_{(21b)}(x, y) = \dot{V}_{(21b)}(z) = 0\}.$$

If E does not contain a whole trajectory of (21b), other than the origin $z = 0$, then

a) the equilibrium $z = 0$ of (21b) is asymptotically stable, and

b) every solution of (21b) starting in H_k at $t = 0$ approaches the origin $z = 0$ as $t \rightarrow \infty$ (i.e., H_k is contained in the domain of attraction of the equilibrium $z = 0$ of (21b)).

The above result is a direct consequence of results given, for e.g., in section 5.11 of Miller and Michel [14]. To obtain the best estimate for the domain of attraction by the use of Theorem 1, for a given V , one chooses the largest value $k = \hat{k}$ for which the hypotheses of this theorem are satisfied. In this case, $H_{\hat{k}}$ is then the desired estimate for the domain of attraction of $z = 0$.

B. Application of the Invariance Principle to Power Systems

In the following, we show that the energy function of the critical group of machines satisfies the hypotheses of Theorem 1. This enables us to deduce the asymptotic stability of the equilibrium $z = 0$ of the entire power system on hand, and it also enables us to obtain an estimate of the domain of attraction of $z = 0$ of the entire power system. We are able to follow this procedure because the energy function V_i for the critical group of machines is dependent on *all* state variables of the power system under discussion.

In the previous sections, we conducted our analysis in the setting of the COI-reference frame. This was done to obtain a tractable expression for energy which causes the critical group of machines to separate from the rest of the system. In order to show that the energy function of the critical group, V_i , is explicitly dependent on *all* state variables, we will conduct our analysis in the following, using the synchronous reference frame (rather than the COI-reference frame). In doing so, the expression for V_i in (12), in terms of the synchronous reference state variables, assumes the form

$$\begin{aligned} V_i = & \frac{1}{2} M_i \left(\omega_i - \frac{1}{M_T} \sum_{j=1}^n M_j \omega_j \right)^2 \\ & - P_i \left(\delta_i - \frac{1}{M_T} \sum_{j=1}^n M_j \delta_j - \delta_i^s + \frac{1}{M_T} \sum_{j=1}^n M_j \delta_j^s \right) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^n C_{ij} \int_{\delta_i^s - \delta_j^s}^{\delta_i - \delta_j} \sin \delta_{ij} d(\delta_i - \delta_j) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^n D_{ij} \int_{\delta_i^s - \delta_j^s}^{\delta_i - \delta_j} \cos \delta_{ij} d(\delta_i - \delta_j) \\ & + \int_{\delta_i^s - \delta_0^s}^{\delta_i - \delta_0} P_{\text{COI}} d(\delta_i - \delta_0), \quad i = 1, 2, \dots, n. \end{aligned} \quad (22)$$

Examining (22) we note that V_i is a function of all $\delta_i, \omega_i, i = 1, 2, \dots, n$. Since the variables θ_i and $\bar{\omega}_i, i = 1, 2, \dots, n$

can be expressed as a linear combination of these variables. V_i satisfies the condition of explicit dependence on all state variables.

Without loss of generality, let machine 1 be the critical machine. We then have the motion of machine 1 governed by

$$\left. \begin{aligned} M_1 \dot{\omega}_1 &= -D_1 \omega_1 + P_1 - P_{e1} \\ \dot{\delta}_1 &= \omega_1 \end{aligned} \right\} \quad (23)$$

and for the noncritical group we have

$$\left. \begin{aligned} M_i \dot{\omega}_i &= -D_i \omega_i + P_i - P_{ei} \\ \dot{\delta}_i &= \omega_i, \quad i = 2, 3, \dots, n. \end{aligned} \right\} \quad (24)$$

Using the notation of (21) we let

$$x = \begin{bmatrix} \omega_1 \\ \delta_1 \end{bmatrix} \quad y = \begin{bmatrix} \omega_2 \\ \delta_2 \\ \omega_3 \\ \delta_3 \\ \vdots \\ \omega_n \\ \delta_n \end{bmatrix} \quad z = \begin{bmatrix} x \\ y \end{bmatrix}. \quad (25)$$

Also, in the sequel we shall use the notation $\omega = (\omega_1, \omega_2, \dots, \omega_n)^T$, and $\delta = (\delta_1, \delta_2, \dots, \delta_n)^T$. This enables us to cast the power system equations into the same form as (21).

Now let $(x(t_{\text{cri}})^T, y(t_{\text{cri}})^T) = z(t_{\text{cri}})^T \triangleq z_{\text{cri}}$ denote the state vector of the entire power system at the instant of critical clearing (t_{cri}) . As we pointed out before (in Section III), it has been shown in several previous works (e.g., [11], [6]–[9]) that $V_1(x, y) = V_1(z) > 0$ for all $z \in H_k^*$, $z \neq 0$, where

$$H_k^* = \{z \in R^{n+m}: V_1(z) \leq V_1(z_{\text{cri}})\}. \quad (26)$$

Since $V_1(z)$ is continuously differentiable, it follows that $V_1(z)$ is positive definite (on H_k^*). It should also be noted that $V_1(z_{\text{cri}}) = V_{\text{PE max}1}(z)$ is obtained along the sustained fault trajectory (see Section IV).

Furthermore, as was shown in Section III, when damping is included in the model, $\dot{V}_1(z) \leq 0$ on this set. Thus hypotheses (i) and (ii) of Theorem 1 are satisfied.

It remains to be shown that the set $E = \{z \in H_k^*: \dot{V}_1(z) = 0\}$ contains no whole trajectory other than the trivial solution $z(t) \equiv 0$. To show this, we argue as follows. Referring to the discussion in Section III, we see that $E = \{z \in H_k^*: \omega(t) \equiv 0\}$. Therefore, if E contains any whole trajectories of (21b), we must have $\omega(t) \equiv 0$. But this implies that (a) $\delta(t) \equiv \text{constant}$, and (b) $\dot{\omega}(t) \equiv 0$. This in turn yields

$$P_i - P_{ei} = 0, \quad i = 1, 2, \dots, n \quad (27)$$

since by assumption $\omega(t) = 0$. But in a power system (described by equation (1)), relation (27) is satisfied if and only if $\delta = \delta^s$ (a stable equilibrium point) or $\delta = \delta^u$ (an unstable equilibrium point). But our region H_k^* excludes the unstable equilibrium point (by the choice of $V_1(z_{\text{cri}})$). Therefore, $z(t) \equiv 0$ is the only whole trajectory in E .

Thus all the hypotheses of Theorem 1 are satisfied and we have shown that: (a) the equilibrium $z = 0$ of the power system described by (21) is asymptotically stable, and (b) the set H_k^* given in (26) is a subset of the domain of attraction (i.e., a subset of the region of stability) of the equilibrium $z = 0$.

Remark

Alternatively, we could have arrived at the same results as the ones presented above (in the present section) by making use of the concept of *partial stability* [15]. In this approach we concern ourselves with the asymptotic behaviour of the $x(t)$ component of the solution $z(t)$, under the assumption that the $y(t)$ component is bounded and tends to the origin as $t \rightarrow \infty$. In this method, we have to use once more an invariance principle [16], which is similar to the one given in Theorem 1.

Even though we are not making explicit use of such results, we have employed in the above analysis the concept of partial stability in a subtle way: given that the noncritical group of machines is asymptotically stable, we can extract more details and obtain sharper estimates in our analysis, since the energy function V_i for the critical group of machines will usually entail fewer constraints than the energy function for the entire power system.

VIII. CONCLUSION

The following conclusions can be drawn from the data and discussion presented in the previous sections.

1) The energy function for an *individual* machine i correctly gives the transient energy in the post-disturbance trajectory, that tends to separate this machine from the rest of the system. Knowing that the noncritical group of machines remains asymptotically stable, it has been shown that, for the *critical group of machines* to remain in synchronism, the post-disturbance stable equilibrium has a region of stability bounded by the hypersurfaces where $V_{\text{PE}i}$ is maximum ($V_{\text{PE}i \text{ max}} = V_{i \text{ critical}}$). A good estimate of $V_{i \text{ critical}}$ is obtained from the sustained fault trajectory.

2) Assessing first swing transient stability by the individual machine energy accurately predicts:

(a) which machines lose synchronism, even in very complex situations, and

(b) critical clearing time (results are the same as those obtained by time resolutions).

3) An analytical justification for using individual machine energy functions to assess transient stability of power systems is provided using an invariance principle. Furthermore, by this invariance principle, a relation between $V_{i \text{ critical}}$ and the domain of attraction has been established.

4) A theory has been proposed which explains the physical phenomenon of power system instability. It precisely determines the mode of instability. It provides less conservative stability results than obtained in previous works which used the *overall* system energy function.

It is to be noted, however, that the present results are preliminary in the sense that the mechanism of separation

of the critical group of machines from the rest of the system needs further investigation.

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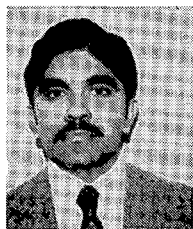
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